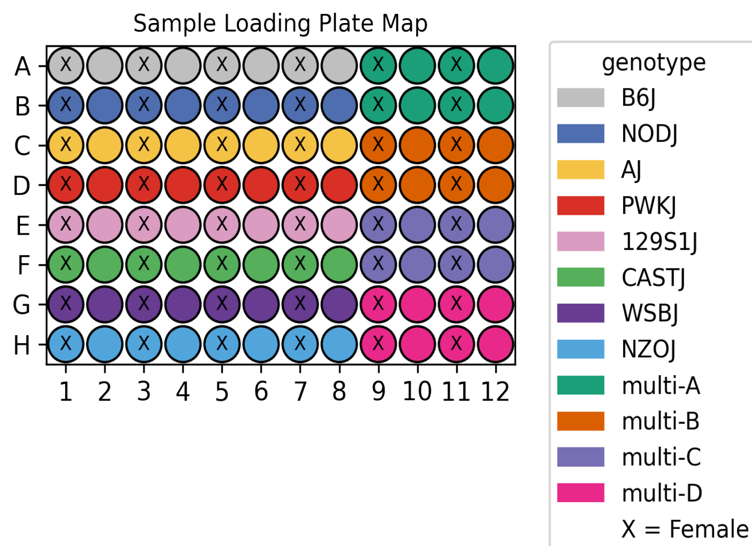
UCI/Caltech IGVF Dataset Pipeline Report

	igvf_010
Kit	WT_mega
Chemistry	Parse v2
Sample type	nuclei
Genotypes	B6J NODJ AJ PWKJ 129S1J CASTJ WSBJ NZOJ
Tissues	Kidney Gastrocnemius
Subpools	16
Exome capture subpools	1
Expected nuclei	1,000,000



1. Sample multiplexing

Kidney was the main tissue on the plate while Gastrocnemius was multiplexed.

Table 1: Genotype multiplexing key

Pair	Multiplexed Genotype 1	Multiplexed Genotype 2
multi-A	B6J	NODJ
multi-B	AJ	PWKJ
multi-C	129S1J	CASTJ
multi-D	NZOJ	WSBJ

2. Alignment

A total of 20,791,665,413 reads were aligned using GENCODE vM32 with mm39 assembly.

Table 2: Pseudoalignment stats

Subpool	Total reads	Total UMIs	Fraction duplicated UMIs	Total pseudoaligned	Pct. pseudoaligned
Subpool_EX	221,200,180	60,766,872	0.68	189,898,163	85.8
Subpool_2	1,276,835,447	483,251,301	0.466	904,757,583	70.9
Subpool_3	1,369,575,547	611,695,187	0.384	993,448,024	72.5
Subpool_4	1,612,033,505	675,704,950	0.422	1,168,793,864	72.5
Subpool_5	1,248,814,638	539,066,620	0.374	861,766,963	69.0
Subpool_6	1,592,999,046	622,581,457	0.453	1,138,170,020	71.4
Subpool_7	1,384,696,974	559,112,645	0.427	974,972,304	70.4
Subpool_8	1,203,983,652	538,471,033	0.381	869,659,557	72.2
Subpool_9	1,466,176,408	660,099,402	0.37	1,047,973,616	71.5
Subpool_10	1,304,338,309	581,250,389	0.377	933,362,412	71.6
Subpool_11	1,425,390,783	621,922,526	0.392	1,022,650,844	71.7
Subpool_12	1,351,956,389	644,918,744	0.339	975,336,573	72.1
Subpool_13	1,350,832,447	656,108,647	0.334	984,510,651	72.9
Subpool_14	1,287,344,112	603,059,517	0.347	923,512,372	71.7
Subpool_15	1,232,967,802	572,778,266	0.354	886,691,022	71.9
Subpool_16	1,462,520,174	648,263,302	0.381	1,046,970,786	71.6

3. Cell recovery

A total of 3,449,552 nuclei across all 16 subpools have >=200 UMIs and 1,161,845 nuclei have >= 500 UMIs.

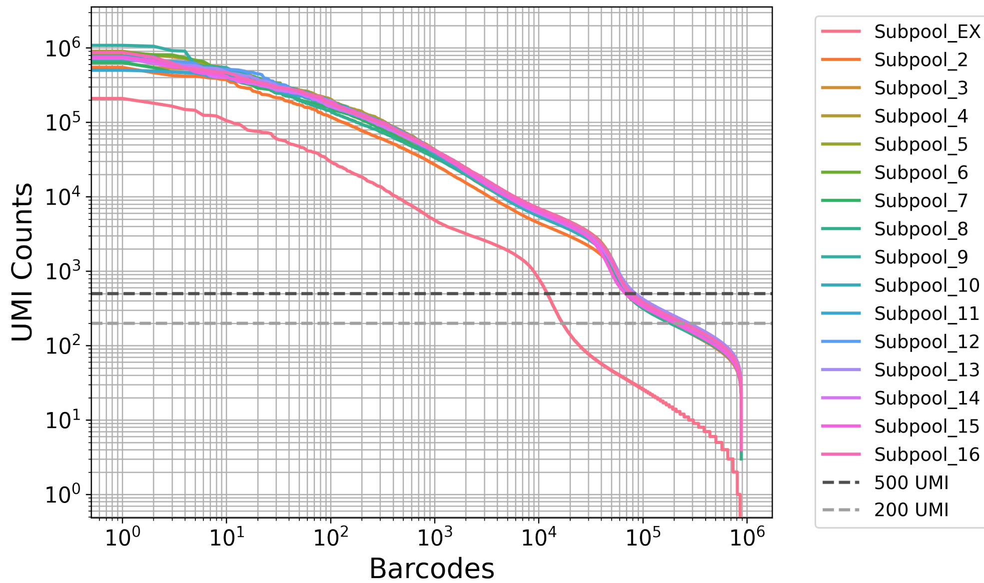


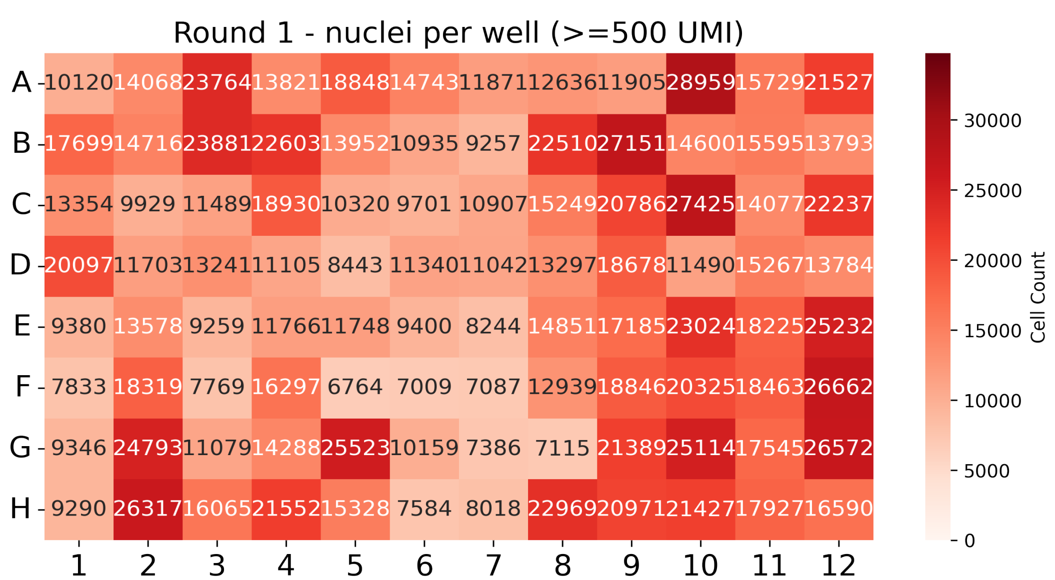
Table 3: QC stats for nuclei >=200 UMI

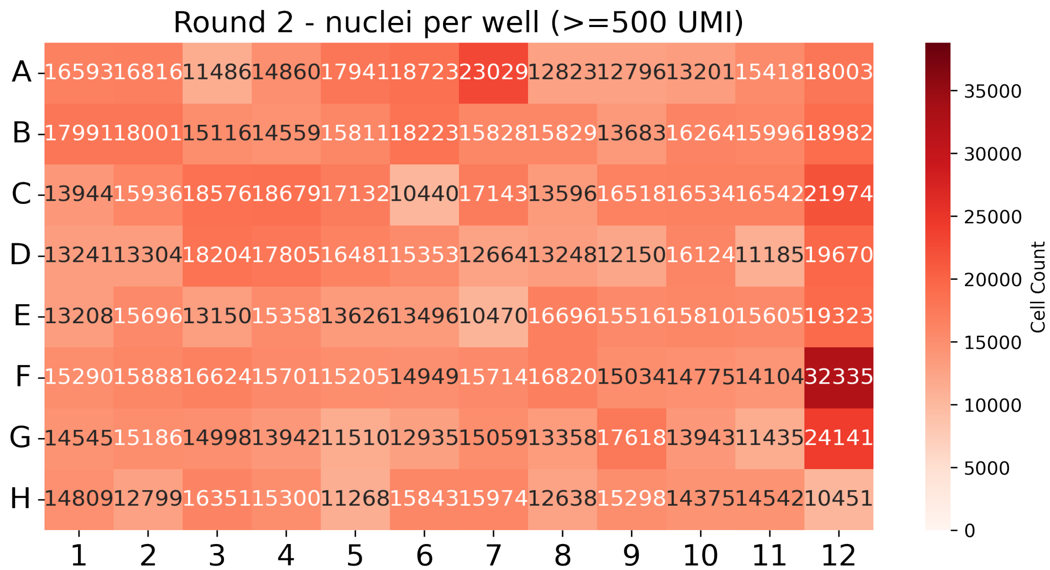
Subpool	Number of nuclei	Median UMIs	Mean UMIs	Median genes	Mean genes
Subpool_EX	16,938	1,109	2,124.9	639	932.1
Subpool_2	191,826	408	1,531.0	305	680.9
Subpool_3	206,737	332	1,956.6	268	735.4
Subpool_4	251,149	326	1,790.9	263	684.7
Subpool_5	185,440	343	1,902.6	275	744.4
Subpool_6	228,993	333	1,770.7	266	686.5
Subpool_7	205,202	342	1,753.9	272	708.9
Subpool_8	184,080	346	1,902.7	273	757.3
Subpool_9	260,731	319	1,666.5	261	653.6
Subpool_10	227,177	321	1,623.2	262	657.7
Subpool_11	250,382	317	1,582.1	259	640.8
Subpool_12	270,602	319	1,600.7	262	637.8
Subpool_13	277,862	323	1,576.5	264	636.7
Subpool_14	239,856	319	1,636.9	260	657.4
Subpool_15	205,525	331	1,861.2	266	704.7
Subpool_16	247,052	324	1,767.5	262	684.7

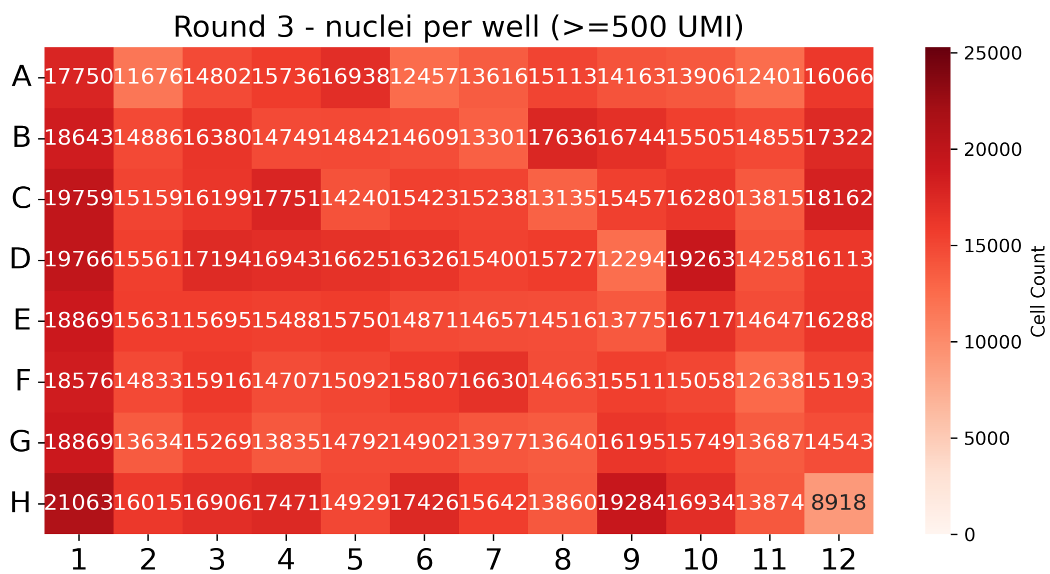
Table 4: QC stats for nuclei >=500 UMI

Subpool	Number of nuclei	Median UMIs	Mean UMIs	Median genes	Mean genes
Subpool_EX	12,105	1,635	2,847.6	893	1,220.8
Subpool_2	82,378	1,589	3,174.9	922	1,278.3
Subpool_3	72,453	2,522	5,056.0	1,279	1,667.4
Subpool_4	80,701	2,407	4,967.2	1,241	1,635.6
Subpool_5	69,473	2,358	4,605.8	1,224	1,601.4
Subpool_6	78,050	2,207	4,640.2	1,167	1,565.2
Subpool_7	74,077	2,232	4,351.4	1,179	1,554.5
Subpool_8	70,128	2,439	4,535.0	1,251	1,617.6
Subpool_9	80,189	2,272	4,775.0	1,194	1,594.8
Subpool_10	74,792	2,121	4,354.1	1,136	1,523.5
Subpool_11	77,415	2,172	4,481.1	1,152	1,550.3
Subpool_12	81,540	2,153	4,646.2	1,154	1,567.4
Subpool_13	85,015	2,099	4,497.4	1,134	1,542.9
Subpool_14	76,270	2,219	4,537.7	1,170	1,567.0
Subpool_15	68,608	2,377	5,002.8	1,238	1,644.5
Subpool_16	78,651	2,416	4,938.4	1,251	1,649.0

In this combinatorial barcoding assay, nuclei are barcoded in 96-well plates over three rounds. The first round assigns sample-specific barcodes, while the next two are applied to the pooled mix. Even distribution in the first round is important to ensure consistent sample representation.







4. Background removal

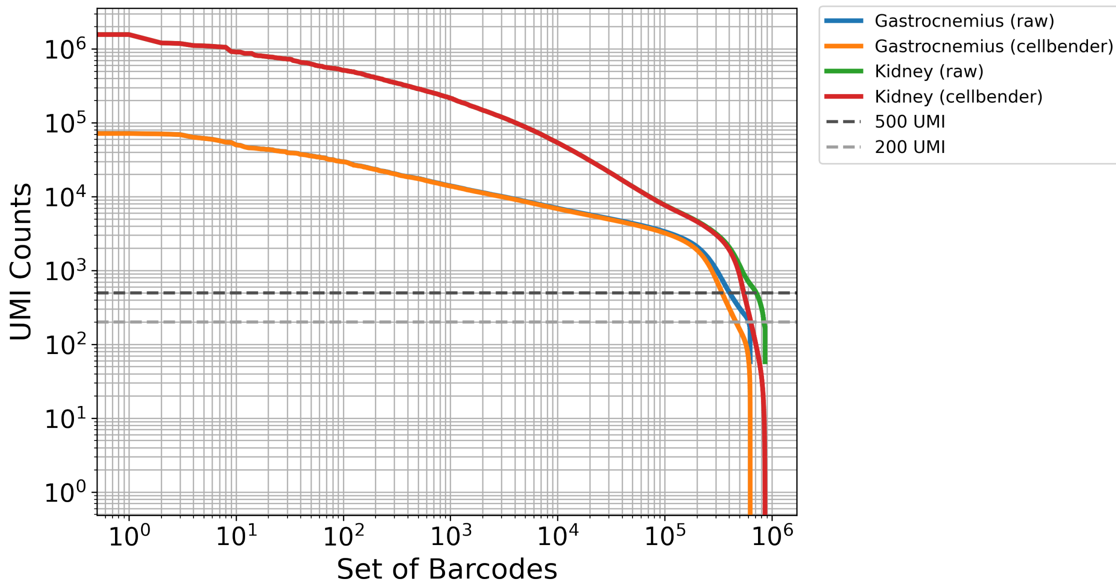
Table 5: Cellbender parameters

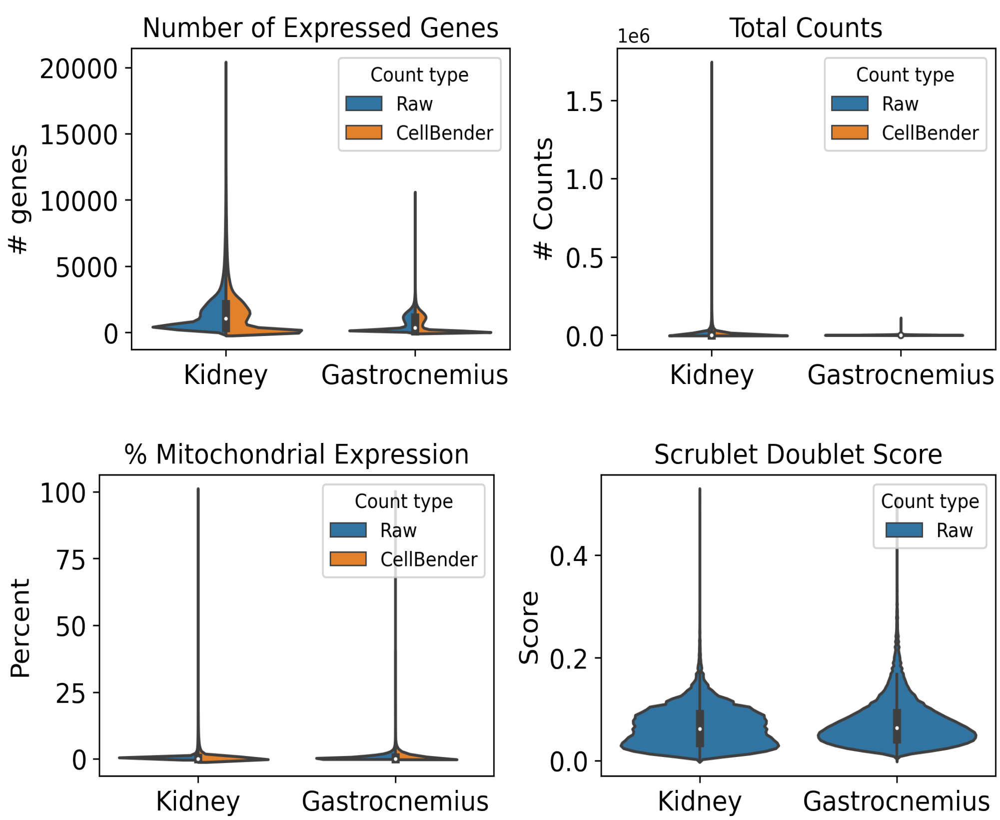
Subpool	droplets_included	learning_rate	expected_cells
Subpool_EX	40000	5e-05	13000
Subpool_2	250000	5e-05	67000
Subpool_3	200000	5e-05	67000
Subpool_4	200000	5e-05	67000
Subpool_5	200000	5e-05	67000

Subpool_6	200000	5e-05	67000
Subpool_7	200000	5e-05	67000
Subpool_8	200000	5e-05	67000
Subpool_9	200000	5e-05	67000
Subpool_10	200000	5e-05	67000
Subpool_11	250000	5e-05	67000
Subpool_12	200000	5e-05	67000
Subpool_13	250000	5e-05	67000
Subpool_14	200000	5e-05	67000
Subpool_15	250000	5e-05	67000
Subpool_16	200000	5e-05	67000

Table 6: Cellbender stats

Subpool	Percent counts removed	Found nuclei	Mean denoised counts
Subpool_EX	3.3	18,853	1,849.3
Subpool_2	8.8	185,577	1,428.6
Subpool_3	5.1	83,458	4,202.3
Subpool_4	5.2	84,450	4,496.5
Subpool_5	5.1	81,910	3,749.8
Subpool_6	5.4	85,594	4,024.8
Subpool_7	6.2	99,985	3,110.6
Subpool_8	4.8	79,029	3,852.2
Subpool_9	6.0	100,614	3,642.6
Subpool_10	5.2	75,578	4,057.9
Subpool_11	5.3	86,554	3,817.8
Subpool_12	5.6	86,293	4,149.3
Subpool_13	6.7	115,393	3,168.8
Subpool_14	6.1	92,649	3,562.1
Subpool_15	6.1	128,635	2,643.9
Subpool_16	5.5	88,554	4,171.0





5. Barcode sample map

The first round corresponds to sample barcoding. In Parse Bio assays, two barcodes are included in the first round: a random hexamer (R) and oligo-dT (T) barcode per well (parse barcode type). This reduces 3-prime bias and allows for the capture of full-length transcripts. These sample level barcodes can be found in the barcode-1 region in the seqspec, and begin at position 16 (0-based) in the full 24-nt cell barcode. For the WT_mega kit, samples are distributed across 96 wells in round 1. The following table includes our lab sample ID as sample descriptions and IGVF sample accessions.

Table 7: Barcode sample map

barcode	sample accession	sample description	position	parse barcode type	well
CATCATCC	IGVFSM0349VWWY	016_B6J_10F_09	16	R	A1
CATTCCTA	IGVFSM0349VWWY	016_B6J_10F_09	16	T	A1
CTGCTTTG	IGVFSM3241ASWH	017_B6J_10M_09	16	R	A2
CTTCATCA	IGVFSM3241ASWH	017_B6J_10M_09	16	T	A2
CTAAGGGA	IGVFSM9474HYEG	018_B6J_10F_09	16	R	A3
CCTATATC	IGVFSM9474HYEG	018_B6J_10F_09	16	T	A3
GCTTATAG	IGVFSM9788LCSW	019_B6J_10M_09	16	R	A4
ACATTTAC	IGVFSM9788LCSW	019_B6J_10M_09	16	T	A4
TCTGATCC	IGVFSM4800CIZG	020_B6J_10F_09	16	R	A5
ACTTAGCT	IGVFSM4800CIZG	020_B6J_10F_09	16	T	A5
TCTCTTGG	IGVFSM7243TNKF	021_B6J_10M_09	16	R	A6
CCAATTCT	IGVFSM7243TNKF	021_B6J_10M_09	16	T	A6
CAATTTCC	IGVFSM8674PTVY	024_B6J_10F_09	16	R	A7
GCCTATCT	IGVFSM8674PTVY	024_B6J_10F_09	16	T	A7
AGTCTCTT	IGVFSM3060AXDG	025_B6J_10M_09	16	R	A8
ATGCTGCT	IGVFSM3060AXDG	025_B6J_10M_09	16	T	A8
TGCTGCTC	IGVFSM4575TTYT	016_B6J_10F_16	16	R	A9
CATTTACA	IGVFSM4575TTYT	016_B6J_10F_16	16	T	A9
TGCTGCTC	IGVFSM4279QZLF	066_NODJ_10F_16	16	R	A9
CATTTACA	IGVFSM4279QZLF	066_NODJ_10F_16	16	T	A9
GTATTTCC	IGVFSM2276CJGW	017_B6J_10M_16	16	R	A10
ACTCGTAA	IGVFSM2276CJGW	017_B6J_10M_16	16	T	A10
GTATTTCC	IGVFSM1252BKWG	067_NODJ_10M_16	16	R	A10
ACTCGTAA	IGVFSM1252BKWG	067_NODJ_10M_16	16	T	A10
TTCCTGTG	IGVFSM0465UALO	018_B6J_10F_16	16	R	A11
CCTTTGCA	IGVFSM0465UALO	018_B6J_10F_16	16	T	A11
TTCCTGTG	IGVFSM8410WIND	074_NODJ_10F_16	16	R	A11

CCTTTGCA	IGVFSM8410WIND	074_NODJ_10F_16	16	T	A11
GCTGCTTC	IGVFSM3666GJLM	019_B6J_10M_16	16	R	A12
ACTCCTGC	IGVFSM3666GJLM	019_B6J_10M_16	16	T	A12
GCTGCTTC	IGVFSM7620ICFV	069_NODJ_10M_16	16	R	A12
ACTCCTGC	IGVFSM7620ICFV	069_NODJ_10M_16	16	T	A12
TATGTGTC	IGVFSM7878IZMN	066_NODJ_10F_09	16	R	B1
ATTTGGCA	IGVFSM7878IZMN	066_NODJ_10F_09	16	T	B1
CAATTCTC	IGVFSM4726ZWYC	067_NODJ_10M_09	16	R	B2
TTATTCTG	IGVFSM4726ZWYC	067_NODJ_10M_09	16	T	B2
TGGTCTCC	IGVFSM4214PFWU	068_NODJ_10F_09	16	R	B3
TCATGCTC	IGVFSM4214PFWU	068_NODJ_10F_09	16	T	B3
GCTCTTTA	IGVFSM8898IMBT	069_NODJ_10M_09	16	R	B4
CATACTTC	IGVFSM8898IMBT	069_NODJ_10M_09	16	T	B4
GCTGCATG	IGVFSM7567RTBV	070_NODJ_10F_09	16	R	B5
CCGTTCTA	IGVFSM7567RTBV	070_NODJ_10F_09	16	T	B5
ACTCATTT	IGVFSM4531FRYY	071_NODJ_10M_09	16	R	B6
GCTTCATA	IGVFSM4531FRYY	071_NODJ_10M_09	16	T	B6
AGTCTTGG	IGVFSM4518CECN	072_NODJ_10F_09	16	R	B7
CTCTGTGC	IGVFSM4518CECN	072_NODJ_10F_09	16	T	B7
GGTTCTTC	IGVFSM4828VZOR	073_NODJ_10M_09	16	R	B8
CCCTTATA	IGVFSM4828VZOR	073_NODJ_10M_09	16	T	B8
TCATGTTG	IGVFSM4471UQDS	020_B6J_10F_16	16	R	B9
ACTGCTCT	IGVFSM4471UQDS	020_B6J_10F_16	16	T	B9
TCATGTTG	IGVFSM3414NFLU	070_NODJ_10F_16	16	R	B9
ACTGCTCT	IGVFSM3414NFLU	070_NODJ_10F_16	16	T	B9
ATTTTGCC	IGVFSM0131WZLT	021_B6J_10M_16	16	R	B10
CTCTAATC	IGVFSM0131WZLT	021_B6J_10M_16	16	T	B10
ATTTTGCC	IGVFSM9301WUUD	071_NODJ_10M_16	16	R	B10
CTCTAATC	IGVFSM9301WUUD	071_NODJ_10M_16	16	T	B10
CTTCTGTA	IGVFSM5168ZHLV	024_B6J_10F_16	16	R	B11
ACCCTTGC	IGVFSM5168ZHLV	024_B6J_10F_16	16	T	B11
CTTCTGTA	IGVFSM5744JVUK	072_NODJ_10F_16	16	R	B11
ACCCTTGC	IGVFSM5744JVUK	072_NODJ_10F_16	16	T	B11
GTCCATCT	IGVFSM2944JZTR	025_B6J_10M_16	16	R	B12
ATCTTAGG	IGVFSM2944JZTR	025_B6J_10M_16	16	T	B12

GTCCATCT	IGVFSM6017WKVH	073_NODJ_10M_16	16	R	B12
ATCTTAGG	IGVFSM6017WKVH	073_NODJ_10M_16	16	T	B12
GCTATCTC	IGVFSM3057IFAI	026_AJ_10F_09	16	R	C1
CATGTCTC	IGVFSM3057IFAI	026_AJ_10F_09	16	T	C1
TAGTTTCC	IGVFSM8586FNQO	029_AJ_10M_09	16	R	C2
TCATTGCA	IGVFSM8586FNQO	029_AJ_10M_09	16	T	C2
TCCATTAT	IGVFSM3536NVMG	028_AJ_10F_09	16	R	C3
ACACCTTT	IGVFSM3536NVMG	028_AJ_10F_09	16	T	C3
AGGATTAA	IGVFSM4523NJGP	031_AJ_10M_09	16	R	C4
AATTTCTC	IGVFSM4523NJGP	031_AJ_10M_09	16	T	C4
AATCTTTC	IGVFSM7352LZKM	030_AJ_10F_09	16	R	C5
ATTCATGG	IGVFSM7352LZKM	030_AJ_10F_09	16	T	C5
GTCATATG	IGVFSM6020DPIE	033_AJ_10M_09	16	R	C6
ACTTTACC	IGVFSM6020DPIE	033_AJ_10M_09	16	T	C6
GTGCTTCC	IGVFSM3951CGSL	032_AJ_10F_09	16	R	C7
CTTCTAAC	IGVFSM3951CGSL	032_AJ_10F_09	16	T	C7
ATGTGTTG	IGVFSM3981QKEE	035_AJ_10M_09	16	R	C8
CTATTTCA	IGVFSM3981QKEE	035_AJ_10M_09	16	T	C8
CCATCTTG	IGVFSM8801AFPS	026_AJ_10F_16	16	R	C9
TCTCATGC	IGVFSM8801AFPS	026_AJ_10F_16	16	T	C9
CCATCTTG	IGVFSM2436WWNM	076_PWKJ_10F_16	16	R	C9
TCTCATGC	IGVFSM2436WWNM	076_PWKJ_10F_16	16	T	C9
TACTGTCT	IGVFSM8141CEOO	027_AJ_10M_16	16	R	C10
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TACTGTCT	IGVFSM8807FWBB	077_PWKJ_10M_16	16	R	C10
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TTCATCGC	IGVFSM4118LWUK	028_AJ_10F_16	16	R	C11
TAAATATC	IGVFSM4118LWUK	028_AJ_10F_16	16	T	C11
TTCATCGC	IGVFSM9269MUDI	078_PWKJ_10F_16	16	R	C11
TAAATATC	IGVFSM9269MUDI	078_PWKJ_10F_16	16	T	C11
ACTGTGGG	IGVFSM2182IYXX	031_AJ_10M_16	16	R	C12
TTACCTGC	IGVFSM2182IYXX	031_AJ_10M_16	16	T	C12
ACTGTGGG	IGVFSM2389EAOO	079_PWKJ_10M_16	16	R	C12
TTACCTGC	IGVFSM2389EAOO	079_PWKJ_10M_16	16	T	C12
TCTGTGCC	IGVFSM9088FTIY	076_PWKJ_10F_09	16	R	D1

CACTTTCA	IGVFSM9088FTIY	076_PWKJ_10F_09	16	T	D1
TCAATCTC	IGVFSM2207LORL	077_PWKJ_10M_09	16	R	D2
CACCTTTA	IGVFSM2207LORL	077_PWKJ_10M_09	16	T	D2
GTCCTCTG	IGVFSM4211TCCR	078_PWKJ_10F_09	16	R	D3
CTGACTTC	IGVFSM4211TCCR	078_PWKJ_10F_09	16	T	D3
TTACATTC	IGVFSM4917CZZF	079_PWKJ_10M_09	16	R	D4
CATTTGGA	IGVFSM4917CZZF	079_PWKJ_10M_09	16	T	D4
ATTCTGTC	IGVFSM4983BATQ	080_PWKJ_10F_09	16	R	D5
GCTCTACT	IGVFSM4983BATQ	080_PWKJ_10F_09	16	T	D5
TGTGTATG	IGVFSM8581GLVP	081_PWKJ_10M_09	16	R	D6
GTTACGTA	IGVFSM8581GLVP	081_PWKJ_10M_09	16	T	D6
TCCATTTG	IGVFSM3357EWRJ	084_PWKJ_10F_09	16	R	D7
CCTGTTGC	IGVFSM3357EWRJ	084_PWKJ_10F_09	16	T	D7
TTAGCTTC	IGVFSM8335WDIG	085_PWKJ_10M_09	16	R	D8
CTATCATC	IGVFSM8335WDIG	085_PWKJ_10M_09	16	T	D8
GTGCTTGA	IGVFSM5510EXRQ	030_AJ_10F_16	16	R	D9
GCTATCAT	IGVFSM5510EXRQ	030_AJ_10F_16	16	T	D9
GTGCTTGA	IGVFSM0334GXDS	080_PWKJ_10F_16	16	R	D9
GCTATCAT	IGVFSM0334GXDS	080_PWKJ_10F_16	16	T	D9
GTTTGTGA	IGVFSM4001HPLV	033_AJ_10M_16	16	R	D10
ACATTCAT	IGVFSM4001HPLV	033_AJ_10M_16	16	T	D10
GTTTGTGA	IGVFSM4237VCZG	081_PWKJ_10M_16	16	R	D10
ACATTCAT	IGVFSM4237VCZG	081_PWKJ_10M_16	16	T	D10
GAAATTAG	IGVFSM2425SGDB	034_AJ_10F_16	16	R	D11
TTCGCTAC	IGVFSM2425SGDB	034_AJ_10F_16	16	T	D11
GAAATTAG	IGVFSM1365IGJT	084_PWKJ_10F_16	16	R	D11
TTCGCTAC	IGVFSM1365IGJT	084_PWKJ_10F_16	16	T	D11
GCAAATTC	IGVFSM7990NFIW	035_AJ_10M_16	16	R	D12
CATTCTAC	IGVFSM7990NFIW	035_AJ_10M_16	16	T	D12
GCAAATTC	IGVFSM1986OPVB	085_PWKJ_10M_16	16	R	D12
CATTCTAC	IGVFSM1986OPVB	085_PWKJ_10M_16	16	T	D12
GAGGTTGA	IGVFSM7015STPL	036_129S1J_10F_09	16	R	E1
CACTTATC	IGVFSM7015STPL	036_129S1J_10F_09	16	T	E1
CCTGTCTG	IGVFSM3870IKYE	037_129S1J_10M_09	16	R	E2
ATAAGCTC	IGVFSM3870IKYE	037_129S1J_10M_09	16	T	E2

GTGGGTTC	IGVFSM2985VTWQ	038_129S1J_10F_09	16	R	E3
TCATCCTG	IGVFSM2985VTWQ	038_129S1J_10F_09	16	T	E3
TTTGCATC	IGVFSM1728JVMJ	041_129S1J_10M_09	16	R	E4
CCTGGTAT	IGVFSM1728JVMJ	041_129S1J_10M_09	16	T	E4
AGGTAATA	IGVFSM7656VCGX	040_129S1J_10F_09	16	R	E5
TGGTATAC	IGVFSM7656VCGX	040_129S1J_10F_09	16	T	E5
GTGCCTTC	IGVFSM3274UVBQ	043_129S1J_10M_09	16	R	E6
TTGGGAGA	IGVFSM3274UVBQ	043_129S1J_10M_09	16	T	E6
ATGTTTCC	IGVFSM9083BJCW	044_129S1J_10F_09	16	R	E7
ACTTCATC	IGVFSM9083BJCW	044_129S1J_10F_09	16	T	E7
CTTAATTC	IGVFSM2309TUXE	045_129S1J_10M_09	16	R	E8
TCTCTAGC	IGVFSM2309TUXE	045_129S1J_10M_09	16	T	E8
TCTGGCTC	IGVFSM6869OUVF	042_129S1J_10F_16	16	R	E9
ATGCCCTT	IGVFSM6869OUVF	042_129S1J_10F_16	16	T	E9
TCTGGCTC	IGVFSM0725SKMI	086_CASTJ_10F_16	16	R	E9
ATGCCCTT	IGVFSM0725SKMI	086_CASTJ_10F_16	16	T	E9
CATCATTT	IGVFSM5187OEBL	037_129S1J_10M_16	16	R	E10
CCCAATTT	IGVFSM5187OEBL	037_129S1J_10M_16	16	T	E10
CATCATTT	IGVFSM8092ISYK	087_CASTJ_10M_16	16	R	E10
CCCAATTT	IGVFSM8092ISYK	087_CASTJ_10M_16	16	T	E10
GTTGTCTC	IGVFSM7768VTDI	038_129S1J_10F_16	16	R	E11
ACTATATA	IGVFSM7768VTDI	038_129S1J_10F_16	16	T	E11
GTTGTCTC	IGVFSM3158KFTN	092_CASTJ_10F_16	16	R	E11
ACTATATA	IGVFSM3158KFTN	092_CASTJ_10F_16	16	T	E11
ATCTTCTG	IGVFSM1059YXHB	041_129S1J_10M_16	16	R	E12
CTCTATAC	IGVFSM1059YXHB	041_129S1J_10M_16	16	T	E12
ATCTTCTG	IGVFSM2176DUOO	089_CASTJ_10M_16	16	R	E12
CTCTATAC	IGVFSM2176DUOO	089_CASTJ_10M_16	16	T	E12
TGTTTGCC	IGVFSM7915MIHZ	086_CASTJ_10F_09	16	R	F1
CTGTCTCA	IGVFSM7915MIHZ	086_CASTJ_10F_09	16	T	F1
TTCTGTCA	IGVFSM1924DVVF	087_CASTJ_10M_09	16	R	F2
GACCTTTC	IGVFSM1924DVVF	087_CASTJ_10M_09	16	T	F2
ACGGACTC	IGVFSM8639WEBM	088_CASTJ_10F_09	16	R	F3
GATTTGGC	IGVFSM8639WEBM	088_CASTJ_10F_09	16	T	F3
TTTGGTCA	IGVFSM0501BSTV	089_CASTJ_10M_09	16	R	F4

CGTCTAGG	IGVFSM0501BSTV	089_CASTJ_10M_09	16	T	F4
TATCCGGG	IGVFSM6781WLWW	090_CASTJ_10F_09	16	R	F5
TACTCGAA	IGVFSM6781WLWW	090_CASTJ_10F_09	16	T	F5
TGTCATTC	IGVFSM7315OJAM	091_CASTJ_10M_09	16	R	F6
CAGCCTTT	IGVFSM7315OJAM	091_CASTJ_10M_09	16	T	F6
ATTCTCTG	IGVFSM3937HVNK	094_CASTJ_10F_09	16	R	F7
CCTCATTA	IGVFSM3937HVNK	094_CASTJ_10F_09	16	T	F7
TGGCTTCC	IGVFSM0131ZKOP	093_CASTJ_10M_09	16	R	F8
CTTATACC	IGVFSM0131ZKOP	093_CASTJ_10M_09	16	T	F8
TTGTTGCC	IGVFSM5139SKXC	040_129S1J_10F_16	16	R	F9
TCTATTAC	IGVFSM5139SKXC	040_129S1J_10F_16	16	T	F9
TTGTTGCC	IGVFSM7844SUGB	090_CASTJ_10F_16	16	R	F9
TCTATTAC	IGVFSM7844SUGB	090_CASTJ_10F_16	16	T	F9
GTCATCTC	IGVFSM7373DCLC	043_129S1J_10M_16	16	R	F10
CCTGCATT	IGVFSM7373DCLC	043_129S1J_10M_16	16	T	F10
GTCATCTC	IGVFSM1128BAML	091_CASTJ_10M_16	16	R	F10
CCTGCATT	IGVFSM1128BAML	091_CASTJ_10M_16	16	T	F10
TTGCTCAT	IGVFSM4259FCLE	044_129S1J_10F_16	16	R	F11
CAATCCTT	IGVFSM4259FCLE	044_129S1J_10F_16	16	T	F11
TTGCTCAT	IGVFSM4593QSDG	094_CASTJ_10F_16	16	R	F11
CAATCCTT	IGVFSM4593QSDG	094_CASTJ_10F_16	16	T	F11
CTGTCTGC	IGVFSM7248MKIA	045_129S1J_10M_16	16	R	F12
TTGTCTTA	IGVFSM7248MKIA	045_129S1J_10M_16	16	T	F12
CTGTCTGC	IGVFSM3110AEWG	093_CASTJ_10M_16	16	R	F12
TTGTCTTA	IGVFSM3110AEWG	093_CASTJ_10M_16	16	T	F12
TATATTCC	IGVFSM9077YRZW	058_WSBJ_10F_09	16	R	G1
TCACTTTA	IGVFSM9077YRZW	058_WSBJ_10F_09	16	T	G1
ATATTGGC	IGVFSM1334FZTF	057_WSBJ_10M_09	16	R	G2
TGCTTGGG	IGVFSM1334FZTF	057_WSBJ_10M_09	16	T	G2
GTGTCCTC	IGVFSM6194HVVO	060_WSBJ_10F_09	16	R	G3
CGCTCATT	IGVFSM6194HVVO	060_WSBJ_10F_09	16	T	G3
ATCTTCAT	IGVFSM7285SRRG	059_WSBJ_10M_09	16	R	G4
GCCTCTAT	IGVFSM7285SRRG	059_WSBJ_10M_09	16	T	G4
CGTGGTTG	IGVFSM2970OEMT	062_WSBJ_10F_09	16	R	G5
GAGCACAA	IGVFSM2970OEMT	062_WSBJ_10F_09	16	T	G5

TTGCATCC	IGVFSM0104SYAP	061_WSBJ_10M_09	16	R	G6
CTCTTAAC	IGVFSM0104SYAP	061_WSBJ_10M_09	16	T	G6
TCTTAATC	IGVFSM9060ZOSA	096_WSBJ_10F_09	16	R	G7
TCTAGGCT	IGVFSM9060ZOSA	096_WSBJ_10F_09	16	T	G7
TGCATTTC	IGVFSM4351BICU	063_WSBJ_10M_09	16	R	G8
AATTCTGC	IGVFSM4351BICU	063_WSBJ_10M_09	16	T	G8
GATGTTTC	IGVFSM5504RVNQ	046_NZOJ_10F_16	16	R	G9
CATTCTCA	IGVFSM5504RVNQ	046_NZOJ_10F_16	16	T	G9
GATGTTTC	IGVFSM0217YUAM	058_WSBJ_10F_16	16	R	G9
CATTCTCA	IGVFSM0217YUAM	058_WSBJ_10F_16	16	T	G9
ATCTTGTC	IGVFSM1584RGVM	047_NZOJ_10M_16	16	R	G10
ACTTGCCT	IGVFSM1584RGVM	047_NZOJ_10M_16	16	T	G10
ATCTTGTC	IGVFSM5571BGFV	057_WSBJ_10M_16	16	R	G10
ACTTGCCT	IGVFSM5571BGFV	057_WSBJ_10M_16	16	T	G10
TCATATTC	IGVFSM8444KPOJ	048_NZOJ_10F_16	16	R	G11
ATCATTGC	IGVFSM8444KPOJ	048_NZOJ_10F_16	16	T	G11
TCATATTC	IGVFSM0943LRWD	064_WSBJ_10F_16	16	R	G11
ATCATTGC	IGVFSM0943LRWD	064_WSBJ_10F_16	16	T	G11
TGGCCTCT	IGVFSM4489MUQD	049_NZOJ_10M_16	16	R	G12
GTTCAACA	IGVFSM4489MUQD	049_NZOJ_10M_16	16	T	G12
TGGCCTCT	IGVFSM8559RKMK	059_WSBJ_10M_16	16	R	G12
GTTCAACA	IGVFSM8559RKMK	059_WSBJ_10M_16	16	T	G12
CGTTGTCT	IGVFSM2869NKDL	046_NZOJ_10F_09	16	R	H1
CCATTTGC	IGVFSM2869NKDL	046_NZOJ_10F_09	16	T	H1
TCTTGTCA	IGVFSM8082YJJT	047_NZOJ_10M_09	16	R	H2
GACTTTGC	IGVFSM8082YJJT	047_NZOJ_10M_09	16	T	H2
TATTCCTG	IGVFSM4505WWVP	048_NZOJ_10F_09	16	R	H3
ATTGGCTC	IGVFSM4505WWVP	048_NZOJ_10F_09	16	T	H3
TCCATGTC	IGVFSM7894QTQT	049_NZOJ_10M_09	16	R	H4
GTGCTAGC	IGVFSM7894QTQT	049_NZOJ_10M_09	16	T	H4
TTGTCATC	IGVFSM9303HUHF	050_NZOJ_10F_09	16	R	H5
CTTTCAAC	IGVFSM9303HUHF	050_NZOJ_10F_09	16	T	H5
ATTTCCTG	IGVFSM0181XWPQ	051_NZOJ_10M_09	16	R	H6
ACTATTGC	IGVFSM0181XWPQ	051_NZOJ_10M_09	16	T	H6
GTGTCTCC	IGVFSM1587OINL	052_NZOJ_10F_09	16	R	H7

ACTGGCTT	IGVFSM1587OINL	052_NZOJ_10F_09	16	T	H7
GTGTGTGT	IGVFSM5770UUHX	053_NZOJ_10M_09	16	R	H8
ATTAGGCT	IGVFSM5770UUHX	053_NZOJ_10M_09	16	T	H8
TATGCTTC	IGVFSM5314ZJEY	050_NZOJ_10F_16	16	R	H9
GCCTTTCA	IGVFSM5314ZJEY	050_NZOJ_10F_16	16	T	H9
TATGCTTC	IGVFSM2242OSDW	062_WSBJ_10F_16	16	R	H9
GCCTTTCA	IGVFSM2242OSDW	062_WSBJ_10F_16	16	T	H9
ATGGTGTT	IGVFSM1469JLYZ	051_NZOJ_10M_16	16	R	H10
ATTCTAGG	IGVFSM1469JLYZ	051_NZOJ_10M_16	16	T	H10
ATGGTGTT	IGVFSM7709RATJ	061_WSBJ_10M_16	16	R	H10
ATTCTAGG	IGVFSM7709RATJ	061_WSBJ_10M_16	16	T	H10
GAATAATG	IGVFSM0037THNT	052_NZOJ_10F_16	16	R	H11
CCTTACAT	IGVFSM0037THNT	052_NZOJ_10F_16	16	T	H11
GAATAATG	IGVFSM4280ZZUH	096_WSBJ_10F_16	16	R	H11
CCTTACAT	IGVFSM4280ZZUH	096_WSBJ_10F_16	16	T	H11
CCTCTGTG	IGVFSM5658GGTC	053_NZOJ_10M_16	16	R	H12
ACATTTGG	IGVFSM5658GGTC	053_NZOJ_10M_16	16	T	H12
CCTCTGTG	IGVFSM5083IADN	063_WSBJ_10M_16	16	R	H12
ACATTTGG	IGVFSM5083IADN	063_WSBJ_10M_16	16	T	H12

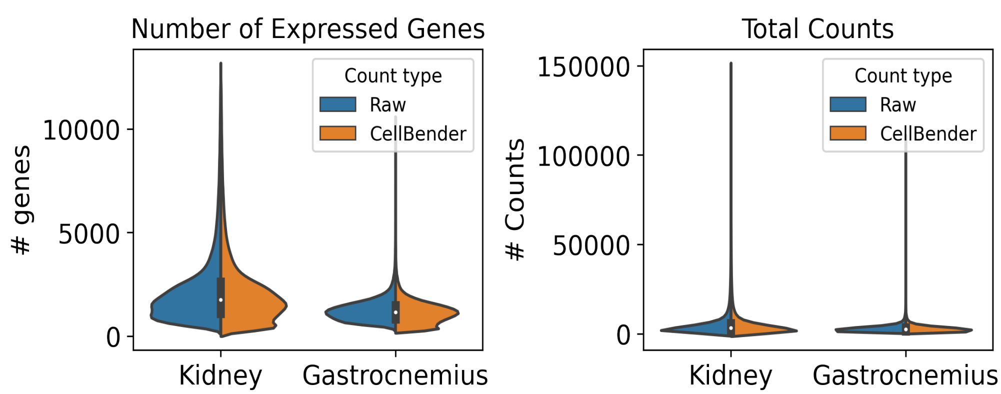
6. Description of data processing workflow and h5ad files

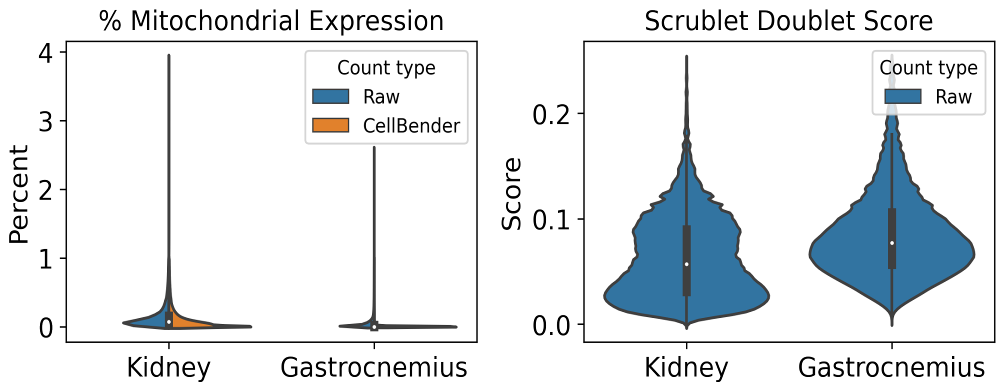
Data from all 16 subpools were concatenated into one AnnData object per tissue and combined with matching data from other plates for processing and cell type annotation. The AnnData contains detailed sample and cell metadata in the observation (obs) table and gene information in the variables table (var). Cell IDs (adata.obs.index) are re-formatted to be human-readable and unique across subpools and experiments by appending subpool and experiment IDs. The final AnnData contains both raw and CellBender denoised counts matrices in separate layers: adata.layers['raw_counts'] and adata.layers['cellbender_counts']. The current adata.X points to the CellBender denoised counts. Cells or nuclei are filtered by >0.5 cell_probability from CellBender analysis.

In addition to CellBender filtering, nuclei were also filtered by the following QC parameters:

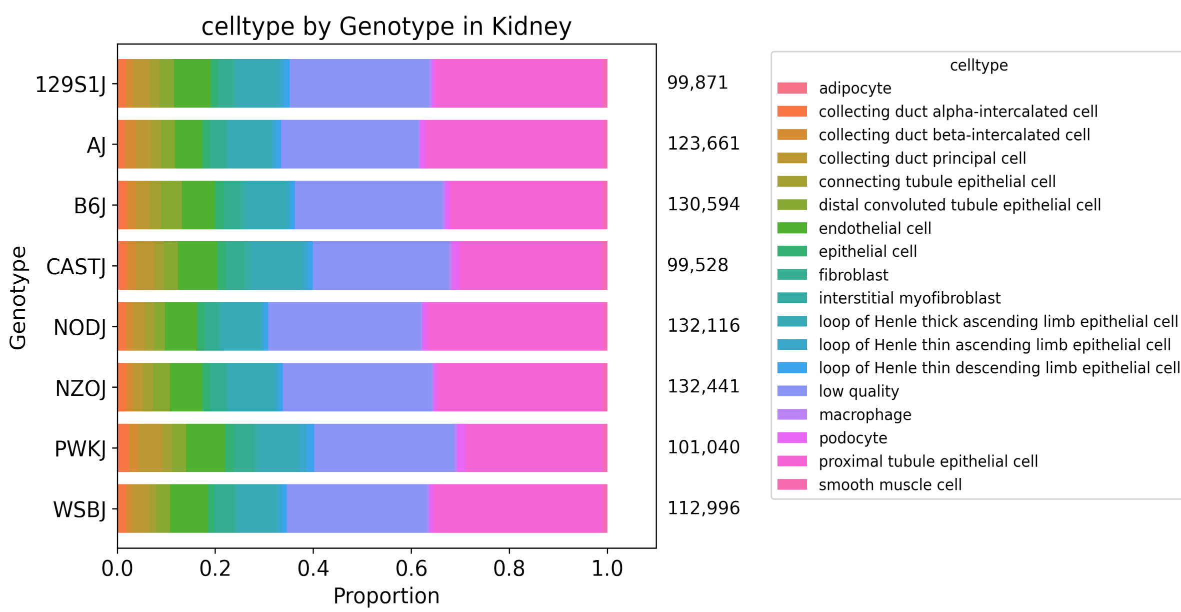
Table 8: QC thresholds

Parameter	Threshold
Minimum # UMIs	500.0
Maximum # UMIs	150000.0
Minimum # genes	250.0
Maximum % mito.	1.0
Maximum doublet score	0.25





After filtering, data for each tissue was normalized and clustered using CellBender denoised counts. Normalization includes regression of technical parameters (number of genes expressed per cell and percent mitochondrial gene expression). Harmony was used to integrate exome capture and non-exome capture subpools for annotation. A Leiden clustering resolution of 1 was used for 54 total clusters in Kidney and 36 clusters in Gastrocnemius, using all tissue data across experiments. Cell type annotations were assigned per Leiden cluster or subcluster (leiden_R) using expression of canonical cell type marker genes.



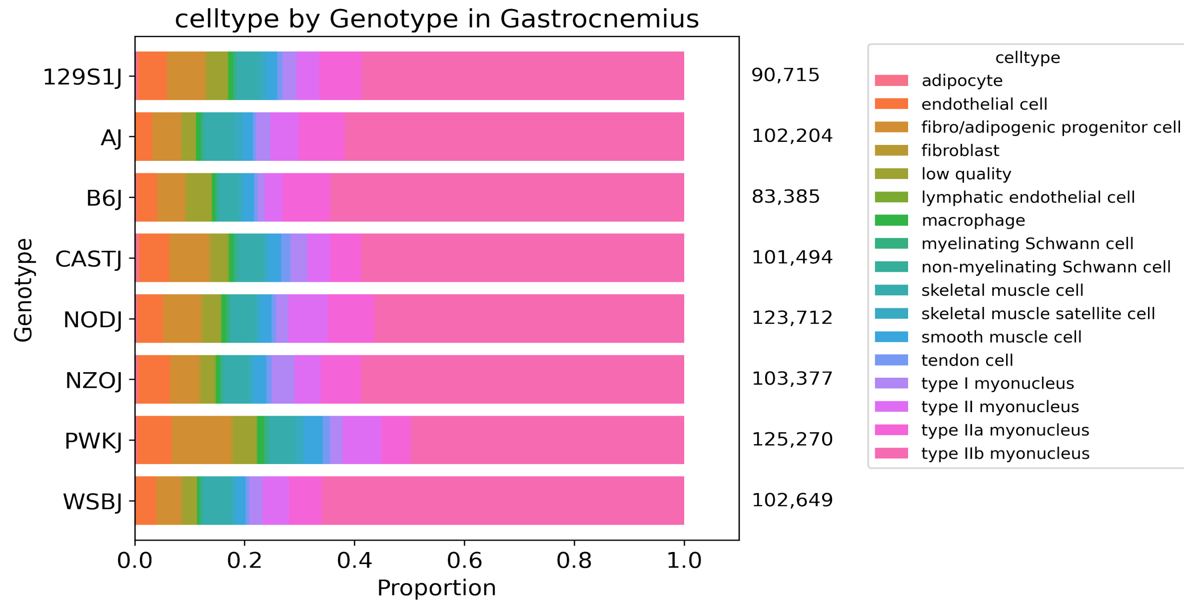


Table 9: Description of obs keys in h5ad file

Key	Description
cellID	Index of obs table. Includes barcoding wells for each round (1_2_3), subpool, and experiment.
lab_sample_id	Human-readable sample name used in-house.
sample	IGVF tissue accession.
plate	Experiment ID, e.g. igvf_003.
subpool	Subpool ID, e.g. Subpool_2. Number corresponds to Illumina index ID from Parse Bio kit.
SampleType	Nuclei or Cells.
Tissue	Human-readable tissue name, e.g. Kidney.
Sex	Male or Female.
Age	Age in postnatal months (PNM) of the mouse, e.g. PNM_02.
Genotype	Mouse genotype/strain, e.g. B6J.
leiden	Numeric Leiden cluster starting from 0.
leiden_R	If necessary, Leiden clusters and any sub-clusters, e.g. 15_1.
subpool_type	Exome capture (EX) or no exome capture (NO).
general_celltype	Broadest level of cell type annotations, e.g. neuron.
general_CL_ID	EMBL CL database ID for general cell type annotation, e.g. CL:0000540.
celltype	Finest level of cell type annotation with a CL ID, e.g. Vip+ GABAergic cortical interneuron.
CL_ID	EMBL CL database ID for cell type annotation, CL:4023016
subtype	Finest level of cell type annotation that may not have a CL ID. Mostly matches celltype.
Protocol	Cell barcoding and library building protocol/kit, e.g. Parse_WT.

Chemistry	Protocol chemistry version, v2 or v3.
bc	24-nucleotide sequence corresponding to 3 8-nucleotide barcode rounds (3, 2, 1).
bc1_sequence	8-nucleotide sequence barcode for round 1.
bc2_sequence	8-nucleotide sequence barcode for round 2.
bc3_sequence	8-nucleotide sequence barcode for round 3.
bc1_well	Round 1 well.
bc2_well	Round 2 well.
bc3_well	Round 2 well.
Row	Sample barcoding plate (round 1) row.
Column	Sample barcoding plate (round 1) column.
well_type	Single or Multiplexed.
Mouse_Tissue_ID	Same as lab_sample_id.
Multiplexed_sample1	Sample 1 from multiplexed well (if applicable)
Multiplexed_sample2	Sample 2 from multiplexed well (if applicable)
DOB	Date of birth of the mouse in month/day/year format.
Age_days	Age of the mouse in days.
Body_weight_g	Body weight in grams of the mouse.
Estrus_cycle	Estimated estrus stage of the mouse, if applicable.
Dissection_date	Date the tissue was harvested in month/day/year format.
Dissection_time	Time in hours:minutes (24-hour time) the mouse was sacrificed.
ZT	Number of hours into the light cycle, from lights on (06:30, ZT0) to lights off (20:30, ZT14).
Dissector	Initials of technician who dissected the tissue.
Tissue_weight_mg	Tissue weight in milligrams.
Notes	Notes taken during sample collection, experiment, or data processing.
n_genes_by_counts_raw	The number of genes with at least 1 count, calculated using raw counts.
total_counts_raw	Total counts (UMIs), calculated using raw counts.
total_counts_mt_raw	Total counts for mitochondrial genes, calculated using raw counts.
pct_counts_mt_raw	Percent of total counts in mitochondrial genes, calculated using raw counts.
n_genes_by_counts_cb	The number of genes with at least 1 count, calculated using CellBender counts.
total_counts_cb	Total counts (UMIs), calculated using CellBender counts.
total_counts_mt_cb	Total counts for mitochondrial genes, calculated using CellBender counts.
pct_counts_mt_cb	Percent of total counts in mitochondrial genes, calculated using CellBender counts.
doublet_score	Scrublet doublet score, calculated using CellBender counts.
predicted_doublet	Scrublet predicted doublet (True or False).
background_fraction	Fraction of counts CellBender predicted to be background noise.

cell_probability	CellBender probability that the cell is real.
cell_size	CellBender size factor.
droplet_efficiency	CellBender statistic indicating transcript capture efficiency per cell.
B6J_klue_counts	Estimated counts for B6J genotype from klue analysis.
NODJ_klue_counts	Estimated counts for NODJ genotype from klue analysis.
AJ_klue_counts	Estimated counts for AJ genotype from klue analysis.
PWKJ_klue_counts	Estimated counts for PWKJ genotype from klue analysis.
129S1J_klue_counts	Estimated counts for 129S1J genotype from klue analysis.
CASTJ_klue_counts	Estimated counts for CASTJ genotype from klue analysis.
WSBJ_klue_counts	Estimated counts for WSBJ genotype from klue analysis.
NZOJ_klue_counts	Estimated counts for NZOJ genotype from klue analysis.

7. Package versions

Table 10: Package versions

package	version
kallisto	0.50.1
bustools	0.43.2
kb	0.28.2
Python	3.9.0
snakemake	7.32.0
cellbender	0.3.2
scanpy	1.10.2
scrublet	0.2.3
anndata	0.10.8
pandas	2.2.2
numpy	1.26.4
harmonypy	0.0.9

8. Contact information and GitHub

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Pipeline repo: https://github.com/mortazavilab/parse_pipeline

Code used to make this report: https://github.com/mortazavilab/8cube_manuscript/blob/main/plate_report/Plate_report_igvf_010.ipynb

Code used to process and annotate the data for main tissue: https://github.com/mortazavilab/8cube_manuscript/blob/main/annotation/Kidney.ipynb

Code used to process and annotate the data for multiplexed tissue: https://github.com/mortazavilab/8cube_manuscript/blob/main/annotation/Gastrocnemius.ipynb